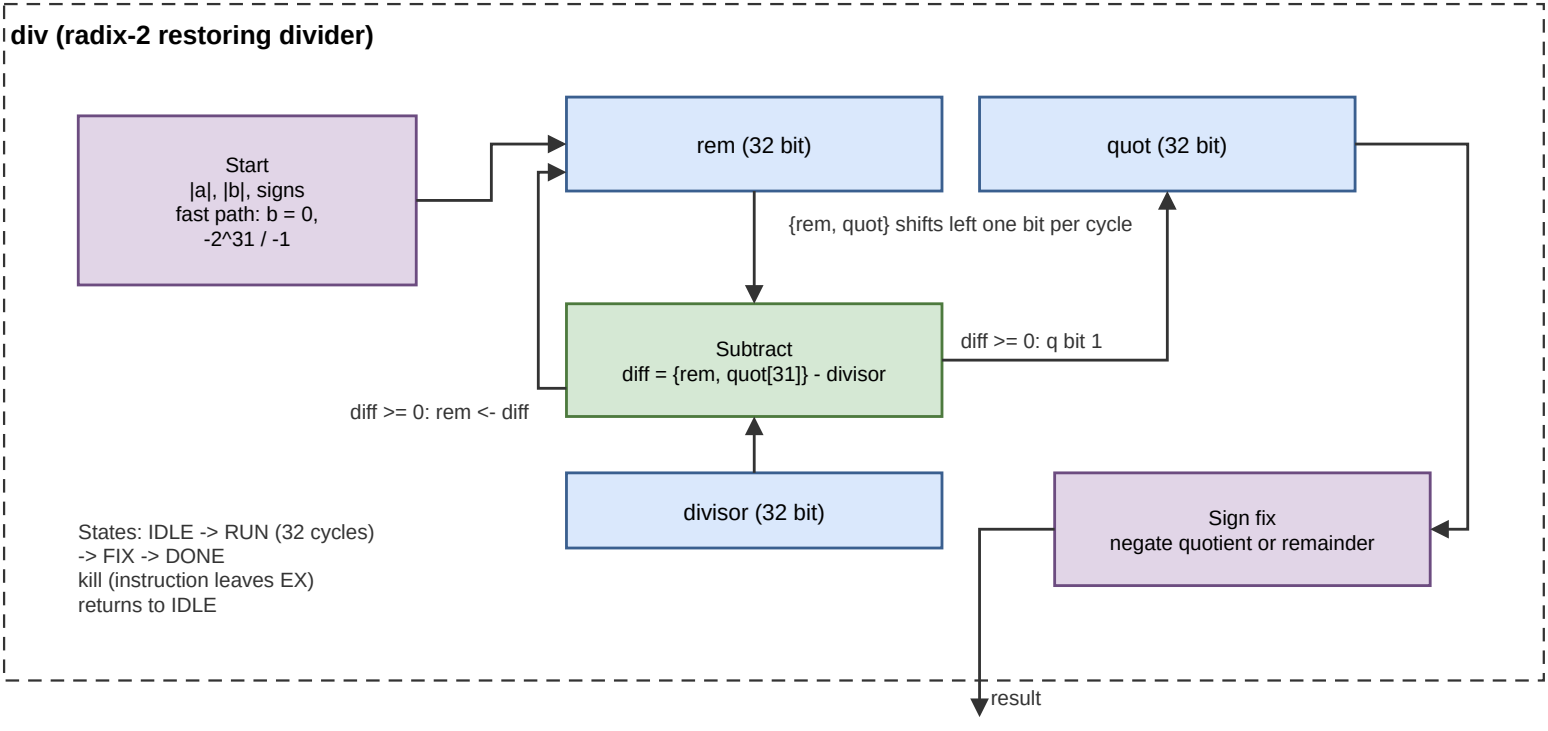
div (radix-2 restoring divider)
Start
|a|, |b|, signs
fast path: b = 0,
-2^31 / -1
rem (32 bit)
quot (32 bit)
{rem, quot} shifts left one bit per cycle
Subtract
diff = {rem, quot[31]} - divisor
diff >= 0: q bit 1
diff >= 0: rem <- diff
divisor (32 bit)
Sign fix
negate quotient or remainder
States: IDLE -> RUN (32 cycles)
-> FIX -> DONE
kill (instruction leaves EX)
returns to IDLE
result